

# $m$ -qh dgas

## Abstract

We formulate quasi-heredity for basic proper connective dg algebras using exceptional standard objects and projective presentations in  $\mathcal{H}_A^m$ . Kernels admit finite filtrations by non-negative shifts of standards with larger weights, with every factor in  $\mathcal{H}_A^m$ . The axioms imply fullness in  $\mathcal{D}_{\text{fd}}(A) = \text{per } A$ , determine the standard objects from the ordered projectives, and place the left-dual objects in  $\mathcal{H}_A^m[1-m]$ . Graded path dg algebras admit such structures for every ordering at a width controlled by path degrees. We obtain an intrinsic criterion for shifted filtrations, an exact least-width formula, and the corresponding injective presentations. The remaining foundational questions concern recognition and comparison of the axioms. We compare replacements for the exceptional-sequence axiom, with counterexamples and a separate summary for each proposal. Finite projective tests can make exceptionality and smoothness consequences; tests using only non-negative shifts fail to control negative cohomology.

## 1 Conventions and definition

Fix an algebraically closed field  $\mathbb{k}$  and integers  $m, n \geq 1$ . All dg modules are right modules, and complexes are cochain complexes:  $d_X : X^q \rightarrow X^{q+1}$  and  $H^q(X[s]) = H^{q+s}(X)$ . Let  $A$  be a proper connective dg algebra with

$$\dim_{\mathbb{k}} H^*(A) < \infty, \quad H^q(A) = 0 \quad (q \notin [1-m, 0]). \quad (1.1)$$

Assume that  $B = H^0(A)$  is basic and that closed degree-zero orthogonal idempotents  $e_1, \dots, e_n$  sum to 1, with classes forming a complete set of primitive idempotents of  $B$ . Use  $e_v$  also for its class in  $B$ , and let  $S_v$  be the simple top of  $e_v B$ . Set

$$\begin{aligned} P_v &= e_v A, & \mathcal{D}_{\text{fd}}(A) &= \{X \in \mathcal{D}(A) : \dim_{\mathbb{k}} H^*(X) < \infty\}, \\ \text{per } A &= \text{thick}_{\mathcal{D}(A)}(A), & \mathcal{H}_A^m &= \mathcal{D}_{\text{fd}}(A)^{[1-m, 0]}. \end{aligned} \quad (1.2)$$

The notation  $\text{thick}$  denotes closure under shifts, cones, and direct summands. Properness gives  $\text{per } A \subseteq \mathcal{D}_{\text{fd}}(A)$ . For a complex  $V$  of vector spaces, write  $\mathbb{D}V = \text{Hom}_{\mathbb{k}}^{\bullet}(V, \mathbb{k})$  for its complex dual.

The category  $\mathcal{H}_A^m$  is extriangulated, with

$$\begin{aligned} \mathbb{E}_{\mathcal{H}_A^m}(X, Y) &= \text{Hom}_{\mathcal{D}_{\text{fd}}(A)}(X, Y[1]), \\ \mathcal{P}(\mathcal{H}_A^m) &= \text{add } A, & \mathcal{I}(\mathcal{H}_A^m) &= \text{add}(\mathbb{D}A[m-1]). \end{aligned} \quad (1.3)$$

Here  $X, Y \in \mathcal{H}_A^m$ ; see [MP25, Corollary 2.3]. A conflation  $K \rightarrow P \rightarrow X \dashrightarrow$  is induced by a triangle  $K \rightarrow P \rightarrow X \rightarrow K[1]$  with all three objects in  $\mathcal{H}_A^m$ . For  $\mathcal{S} \subseteq \mathcal{H}_A^m$ , write  $\text{Filt}_{\mathcal{H}_A^m}(\mathcal{S})$  for the objects  $M$  admitting a finite sequence of conflations

$$M_{a-1} \longrightarrow M_a \longrightarrow F_a \dashrightarrow, \quad M_0 = 0, \quad M_{\ell} = M, \quad F_a \in \mathcal{S} \quad (1 \leq a \leq \ell). \quad (1.4)$$

All  $M_a$  belong to  $\mathcal{H}_A^m$ . Empty filtrations are allowed, and finite direct sums belong to  $\text{Filt}_{\mathcal{H}_A^m}(\mathcal{S})$ ; closure under direct summands is not imposed. The inflations in (1.4) need not be monomorphisms of dg modules.

An object  $E \in \mathcal{D}_{\text{fd}}(A)$  is *exceptional* if

$$\text{Hom}_{\mathcal{D}_{\text{fd}}(A)}(E, E[t]) \cong \begin{cases} \mathbb{k}, & t = 0, \\ 0, & t \neq 0 \end{cases} \quad (t \in \mathbb{Z}). \quad (1.5)$$

An ordered tuple  $(E_1, \dots, E_n)$  of exceptional objects is an *exceptional sequence* if

$$\text{Hom}_{\mathcal{D}_{\text{fd}}(A)}(E_i, E_j[t]) = 0 \quad (1 \leq j < i \leq n, t \in \mathbb{Z}). \quad (1.6)$$

The indices  $i, j$  in (1.6) denote positions in the sequence. For exceptional objects  $\Delta_1, \dots, \Delta_n$ , a permutation  $\sigma \in S_n$  lists the vertex labels in the standard sequence

$$\Delta^\sigma = (\Delta_{\sigma(1)}, \dots, \Delta_{\sigma(n)}). \quad (1.7)$$

Thus  $\sigma^{-1}(v)$  is the position of  $\Delta_v$  in  $\Delta^\sigma$ . Define the total order  $\leq_\sigma$  on  $\{1, \dots, n\}$  by

$$u \leq_\sigma v \iff \sigma^{-1}(u) \leq \sigma^{-1}(v). \quad (1.8)$$

Write  $u <_\sigma v$  if  $u \leq_\sigma v$  and  $u \neq v$ . The least and greatest weights are  $\sigma(1)$  and  $\sigma(n)$ , respectively. Thus smaller weights occur earlier in (1.7). By (1.6), the sequence  $\Delta^\sigma$  is exceptional precisely when

$$\text{Hom}_{\mathcal{D}_{\text{fd}}(A)}(\Delta_w, \Delta_v[t]) = 0 \quad (v <_\sigma w, t \in \mathbb{Z}). \quad (1.9)$$

The sequence  $\Delta^\sigma$  is *full in  $\mathcal{D}_{\text{fd}}(A)$*  if its terms thickly generate  $\mathcal{D}_{\text{fd}}(A)$ .

**Definition 1.1** (*m-qh dga*). An algebra  $A$  satisfying (1.1) is an *m-qh dga* with respect to  $\sigma$  if there are objects  $\Delta_v \in \mathcal{H}_A^m$  such that:

(Q1)  $\Delta^\sigma$  is an exceptional sequence;

(Q2) for every vertex  $v$ , there is a conflation

$$K_v \longrightarrow P_v \xrightarrow{p_v} \Delta_v \dashrightarrow \quad \text{in } \mathcal{H}_A^m, \quad (1.10)$$

with

$$\begin{aligned} K_v &\in \text{Filt}_{\mathcal{H}_A^m}(\mathcal{S}_{>v}^m), \\ \mathcal{S}_{>v}^m &= \{\Delta_w[s] : v <_\sigma w, 0 \leq s < m, \Delta_w[s] \in \mathcal{H}_A^m\}. \end{aligned} \quad (1.11)$$

The shifts in (1.11) are cohomological. The membership  $\Delta_w[s] \in \mathcal{H}_A^m$  is required for each factor, not only for  $K_v$ . For  $v = \sigma(n)$ , the set  $\mathcal{S}_{>v}^m$  is empty, so  $K_v = 0$  and  $p_v : P_v \xrightarrow{\sim} \Delta_v$ . A particular filtration (1.4) is a witness for (1.11), rather than additional chosen structure.

## 2 Consequences of the axioms

Throughout this section, fix a structure satisfying Definition 1.1.

**Proposition 2.1** (Perfection and fullness). *Conditions Q1 and Q2 imply*

$$\Delta_v \in \text{per } A \quad (1 \leq v \leq n), \quad \text{thick}_{\mathcal{D}_{\text{fd}}(A)}(\Delta_1, \dots, \Delta_n) = \text{per } A = \mathcal{D}_{\text{fd}}(A). \quad (2.1)$$

*Proof. Step 1: generation of per A.* Starting with  $\Delta_{\sigma(n)} \cong P_{\sigma(n)}$ , descending induction in  $\leq_\sigma$  and (1.11) show that  $K_v$  and  $\Delta_v$  are perfect. Conversely, (1.10) and (1.11) put every  $P_v$  in  $\text{thick}_{\mathcal{D}_{\text{fd}}(A)}(\Delta_1, \dots, \Delta_n)$ .

*Step 2: generation of  $\mathcal{D}_{\text{fd}}(A)$ .* For  $X \in \mathcal{D}_{\text{fd}}(A)$ , put  $X_n = X$ . For  $a = n, \dots, 1$ , take the evaluation triangle

$$\begin{aligned} V_a &= \bigoplus_{t \in \mathbb{Z}} \text{Hom}_{\mathcal{D}_{\text{fd}}(A)}(\Delta_{\sigma(a)}, X_a[t])[-t], \\ V_a \otimes_{\mathbb{k}} \Delta_{\sigma(a)} &\xrightarrow{\text{ev}_a} X_a \longrightarrow X_{a-1} \longrightarrow (V_a \otimes_{\mathbb{k}} \Delta_{\sigma(a)})[1]. \end{aligned} \quad (2.2)$$

Each  $V_a$  is a complex with zero differential. The sums are finite: the first argument is perfect, and the second has finite-dimensional total cohomology. Exceptionality gives  $\text{Hom}_{\mathcal{D}_{\text{fd}}(A)}(\Delta_{\sigma(a)}, X_{a-1}[t]) = 0$  for all  $t$ ; (1.9) preserves the vanishings for positions  $b > a$ . Thus  $\text{Hom}_{\mathcal{D}_{\text{fd}}(A)}(P_v, X_0[t]) = 0$  for every  $v, t$ . Since  $\text{Hom}_{\mathcal{D}_{\text{fd}}(A)}(P_v, X_0[t]) = H^t(X_0)e_v$ , we obtain  $X_0 = 0$ . The triangles (2.2) now express  $X$  in  $\text{thick}_{\mathcal{D}_{\text{fd}}(A)}(\Delta_1, \dots, \Delta_n)$ .  $\square$

**Corollary 2.2** (Smoothness). *Every algebra satisfying Definition 1.1 is homologically smooth: its diagonal bimodule belongs to  $\text{per}(A^{\text{op}} \otimes_{\mathbb{k}} A)$ .*

*Proof.* Proposition 2.1 makes every simple  $B$ -module  $S_v$  perfect over  $A$ . Since  $B$  is split basic,  $B/\text{rad } B \cong \mathbb{k}^n$  is separable over  $\mathbb{k}$ . The smoothness criterion [RS22, Theorem 5.8] applies.  $\square$

The *left-dual sequence* of  $\Delta^\sigma$  is the exceptional sequence  $(\nabla_{\sigma(n)}, \dots, \nabla_{\sigma(1)})$ , listed in decreasing order for  $\leq_\sigma$ , with terms in  $\text{thick}_{\mathcal{D}_{\text{fd}}(A)}(\Delta_1, \dots, \Delta_n)$ , satisfying

$$\text{Hom}_{\mathcal{D}_{\text{fd}}(A)}(\Delta_u, \nabla_v[t]) \cong \begin{cases} \mathbb{k}, & u = v, t = 0, \\ 0, & \text{otherwise} \end{cases} \quad (1 \leq u, v \leq n, t \in \mathbb{Z}). \quad (2.3)$$

This is the convention of [BB26, §2.3].

**Proposition 2.3** (Left-dual objects). *The sequence  $\Delta^\sigma$  has a unique left-dual sequence up to isomorphism of its terms. Its objects satisfy*

$$\nabla_v \in \mathcal{H}_A^m[1 - m], \quad \mathcal{H}_A^m[1 - m] = \mathcal{D}_{\text{fd}}(A)^{[0, m-1]}. \quad (2.4)$$

*Proof.* Perfection and properness make all mutation triangles finite. For  $v = \sigma(a)$ , successively take evaluation cones of  $\Delta_v$  against  $\Delta_{\sigma(a-1)}, \dots, \Delta_{\sigma(1)}$ ; for  $a = 1$  take  $\nabla_{\sigma(1)} = \Delta_{\sigma(1)}$ . Semiorthogonality gives (2.3), and the mutation construction gives the reversed exceptional sequence  $(\nabla_{\sigma(n)}, \dots, \nabla_{\sigma(1)})$  [BB26, §2.3]. Proposition 2.1 and the characterisation by (2.3) for a full exceptional sequence give uniqueness [BB26, §2.3].

For every factor  $\Delta_w[s] \in \mathcal{S}_{>v}^m$ ,

$$\text{Hom}_{\mathcal{D}_{\text{fd}}(A)}(\Delta_w[s], \nabla_u[t]) \cong \begin{cases} \mathbb{k}, & w = u, t = s, \\ 0, & \text{otherwise.} \end{cases} \quad (2.5)$$

The long exact sequences from the filtrations of  $K_v$  therefore give  $\text{Hom}_{\mathcal{D}_{\text{fd}}(A)}(K_v, \nabla_u[t]) = 0$  for  $t \notin [0, m-1]$ . Applying  $\text{Hom}_{\mathcal{D}_{\text{fd}}(A)}(-, \nabla_u[t])$  to (1.10) gives

$$H^t(\nabla_u)e_v = \text{Hom}_{\mathcal{D}_{\text{fd}}(A)}(P_v, \nabla_u[t]) = 0 \quad (t \notin [0, m-1]).$$

Summing over  $v$  proves (2.4).  $\square$

The fullness of  $\Delta^\sigma$ , existence of its left-dual sequence  $(\nabla_{\sigma(n)}, \dots, \nabla_{\sigma(1)})$ , and (2.4) are consequences of Definition 1.1.

**Proposition 2.4** (Standards determined by the ordered projectives). *For  $1 \leq v \leq n$ , put*

$$\mathcal{Q}_v = \text{thick}_{\mathcal{D}_{\text{fd}}(A)}\{P_w : v <_\sigma w\}, \quad e_{>v}^\sigma = \sum_{v <_\sigma w} e_w. \quad (2.6)$$

Write  $\mathcal{Q}_v^\perp$  for the objects  $X \in \mathcal{D}_{\text{fd}}(A)$  such that  $\text{Hom}_{\mathcal{D}_{\text{fd}}(A)}(Y, X[t]) = 0$  for every  $Y \in \mathcal{Q}_v$  and  $t \in \mathbb{Z}$ . Then

$$\begin{aligned} K_v &\in \mathcal{Q}_v, & \Delta_v &\in \mathcal{Q}_v^\perp, \\ p_v^* : \text{Hom}_{\mathcal{D}_{\text{fd}}(A)}(\Delta_v, X[t]) &\xrightarrow{\sim} \text{Hom}_{\mathcal{D}_{\text{fd}}(A)}(P_v, X[t]) & (X \in \mathcal{Q}_v^\perp, t \in \mathbb{Z}). \end{aligned} \quad (2.7)$$

Consequently  $(\Delta_v, p_v)$  is determined by  $(A, \sigma)$  up to a unique isomorphism compatible with  $p_v$ , and  $K_v$  is determined up to isomorphism. Moreover,

$$\begin{aligned} H^t(\Delta_v)e_w &= 0 \quad (v <_\sigma w, t \in \mathbb{Z}), \\ H^t(\Delta_v)e_v &\cong \begin{cases} \mathbb{k}, & t = 0, \\ 0, & t \neq 0, \end{cases} \quad H^0(\Delta_v) \cong e_v B / e_v B e_{>v}^\sigma B. \end{aligned} \quad (2.8)$$

The morphism  $H^0(p_v) : e_v B \rightarrow H^0(\Delta_v)$  is a projective cover with simple top  $S_v$ .

*Proof.* The induction in Step 1 of Proposition 2.1, restricted to weights  $w >_\sigma v$ , gives  $\mathcal{Q}_v = \text{thick}_{\mathcal{D}_{\text{fd}}(A)}\{\Delta_w : v <_\sigma w\}$ . Equations (1.11) and (1.9) imply  $K_v \in \mathcal{Q}_v$  and  $\Delta_v \in \mathcal{Q}_v^\perp$ . Applying  $\text{Hom}_{\mathcal{D}_{\text{fd}}(A)}(-, X[t])$  to (1.10) proves (2.7) and its uniqueness assertion. Taking  $X = \Delta_v$  gives  $\text{Hom}_{\mathcal{D}_{\text{fd}}(A)}(P_v, \Delta_v[t]) \cong \text{Hom}_{\mathcal{D}_{\text{fd}}(A)}(\Delta_v, \Delta_v[t])$ ; the other vertex vanishings follow from  $\Delta_v \in \mathcal{Q}_v^\perp$ .

For the degree-zero formula, set  $D_v^0 = e_v B / e_v B e_{>v}^\sigma B$ . The module  $D_v^0$  belongs to  $\mathcal{Q}_v^\perp$ , so the quotient  $P_v \rightarrow D_v^0$  factors through  $p_v$  by (2.7). Conversely, the support vanishings and the epimorphism  $H^0(p_v)$  give a factorisation  $e_v B \rightarrow D_v^0 \rightarrow H^0(\Delta_v)$ . The two induced maps between  $D_v^0$  and  $H^0(\Delta_v)$  are inverse, as can be checked after precomposition with their epimorphisms from  $e_v B$ . Finally,  $H^0(\Delta_v)e_v \cong \mathbb{k}$ , so  $H^0(\Delta_v)$  is a non-zero quotient of the indecomposable projective  $e_v B$ . Thus  $H^0(p_v)$  is a projective cover with top  $S_v$ .  $\square$

**Corollary 2.5** (Classical specialisation). *For  $m = 1$ , Definition 1.1 is equivalent to classical quasi-heredity of  $B = H^0(A)$  with respect to  $\leq_\sigma$ .*

*Proof.* The standard  $t$ -structure identifies  $\mathcal{H}_A^1$  with  $\text{mod } B$ , and  $A$  is quasi-isomorphic to  $B$  under (1.1) with  $m = 1$ .

Shifted axioms  $\Rightarrow$  classical axioms. Only  $s = 0$  occurs in (1.11). The standard objects are the usual standard modules by (2.8), and (1.10) becomes a short exact sequence whose kernel is filtered by standards of larger weights. These are the classical projective-filtration conditions [Kra17, §3].

Classical axioms  $\Rightarrow$  shifted axioms. Classical standard modules satisfy (1.10) and (1.11) with  $s = 0$ . They form an exceptional sequence [BB26, Lemma 2.2], so condition Q1 holds.  $\square$

**Example 2.6** (The order for two classical standards). Take  $A = B = \mathbb{k}(1 \xrightarrow{a} 2)$  concentrated in degree zero, with  $e_1 a = a = a e_2$ ,  $m = 1$ , and  $\sigma = \text{id}$ . Thus  $1 <_\sigma 2$ . Let  $S_v$  be the simple top of  $P_v = e_v B$ , for  $v = 1, 2$ . Then

$$\Delta_1 = S_1, \quad \Delta_2 = S_2 = P_2, \quad \nabla_1 = S_1, \quad \nabla_2 = P_1. \quad (2.9)$$

The projective presentation

$$0 \longrightarrow S_2 \longrightarrow P_1 \longrightarrow S_1 \longrightarrow 0 \quad (2.10)$$

gives

$$\text{Hom}_{\mathcal{D}_{\text{fd}}(B)}(S_2, S_1[t]) = 0 \quad (t \in \mathbb{Z}), \quad \text{Hom}_{\mathcal{D}_{\text{fd}}(B)}(S_1, S_2[1]) \cong \mathbb{k}. \quad (2.11)$$

Consequently  $(\Delta_1, \Delta_2) = (S_1, S_2)$  is full exceptional, whereas  $(\Delta_2, \Delta_1)$  is not exceptional. The left-dual sequence of  $(\Delta_1, \Delta_2)$  specified by (2.3) is  $(\nabla_2, \nabla_1) = (P_1, S_1)$ .

### 3 Graded path dg algebras

Let  $Q$  be a finite acyclic quiver on  $\{1, \dots, n\}$ , with all arrow degrees at most zero. Paths are concatenated from left to right:  $e_u a = a = a e_v$  for an arrow  $a : u \rightarrow v$ . Let  $A = (\mathbb{k}Q, d_A)$ , where  $d_A$  is any graded derivation of degree 1 with  $d_A^2 = 0$ . All vertex idempotents are closed. For a path  $p$ , let  $|p|$  denote its degree, including  $|e_v| = 0$ , and put

$$M = 1 - \min\{|p| : p \text{ is a path in } Q\}. \quad (3.1)$$

Thus the underlying graded algebra is concentrated in degrees  $[1 - M, 0]$ .

**Theorem 3.1** (Every ordering). *For every  $\sigma \in S_n$ , define*

$$B_v = A/Ae_{>v}^\sigma A, \quad \Delta_v = e_v B_v, \quad K_v = e_v A e_{>v}^\sigma A. \quad (3.2)$$

*The objects (3.2) give an  $M$ -qh dga. The same presentations and filtrations give an  $m$ -qh dga for every  $m \geq M$ .*

*Proof. Step 1: the kernel filtration.* Fix  $v = \sigma(i)$ . If  $i = n$ , then  $K_v = 0$  and its filtration is empty. For  $i < n$ , define dg ideals and dg submodules by

$$J_j = A \left( \sum_{b=j}^n e_{\sigma(b)} \right) A, \quad F_j = e_v J_j, \quad J_{n+1} = 0, \quad (3.3)$$

$$0 = F_{n+1} \subseteq F_n \subseteq \cdots \subseteq F_{i+1} = K_v.$$

For  $i < j \leq n$  and  $w = \sigma(j)$ , put  $L_{vw} = e_v B_w e_w$ , a finite complex of vector spaces. Multiplication gives a dg isomorphism

$$L_{vw} \otimes_{\mathbb{k}} \Delta_w \xrightarrow{\sim} F_j/F_{j+1}, \quad p \otimes q \mapsto pq. \quad (3.4)$$

Indeed, a basis path in  $F_j/F_{j+1}$  has greatest vertex  $w$  in  $\leq_\sigma$ . Acyclicity permits a unique cut at the sole occurrence of  $w$ . The Leibniz rule makes (3.4) a morphism of complexes. Splitting  $L_{vw}$  over  $\mathbb{k}$  gives

$$F_j/F_{j+1} \simeq \bigoplus_{h=1-M}^0 \Delta_w[-h]^{\oplus \dim_{\mathbb{k}} H^h(L_{vw})} \quad \text{in } \mathcal{D}_{\text{fd}}(A). \quad (3.5)$$

*Step 2: every factor stays in  $\mathcal{H}_A^M$ .* If  $H^h(L_{vw}) \neq 0$ , there is a prefix path  $p : v \rightarrow w$  of degree  $h$ . For every basis path  $q$  of  $\Delta_w$ , the concatenation  $pq$  is a path, and  $h + |q| \geq 1 - M$ . Thus  $\Delta_w[-h] \in \mathcal{H}_A^M$  and  $0 \leq -h < M$ . All submodules and quotients in (3.3) are themselves concentrated in  $[1 - M, 0]$ . Their short exact sequences induce conflations in  $\mathcal{H}_A^M$ ; refining (3.5) gives (1.11). The short exact sequence  $0 \rightarrow K_v \rightarrow P_v \rightarrow \Delta_v \rightarrow 0$  gives (1.10).

*Step 3: exceptionality.* Fix  $v$ . The path basis gives  $\Delta_v e_w = 0$  for  $v <_\sigma w$  and  $\Delta_v e_v = \mathbb{k}$  in degree zero. Descending induction on  $w >_\sigma v$ , using the presentations of  $\Delta_w$  and the filtrations of  $K_w$ , gives  $\text{Hom}_{\mathcal{D}_{\text{fd}}(A)}(\Delta_w, \Delta_v[t]) = 0$  for every  $t$ . The filtration of  $K_v$  then gives  $\text{Hom}_{\mathcal{D}_{\text{fd}}(A)}(K_v, \Delta_v[t]) = 0$ . Consequently

$$\text{Hom}_{\mathcal{D}_{\text{fd}}(A)}(\Delta_v, \Delta_v[t]) \cong \text{Hom}_{\mathcal{D}_{\text{fd}}(A)}(P_v, \Delta_v[t]) = H^t(\Delta_v e_v),$$

which is  $\mathbb{k}$  for  $t = 0$  and zero otherwise. For  $m \geq M$ , every object in the presentations and filtrations still belongs to  $\mathcal{H}_A^m$ , and every factor shift still satisfies  $0 \leq s < m$ .  $\square$

**Example 3.2** (A graded quiver of type  $A_4$ ). Take  $Q : 1 \xrightarrow{a} 2 \xrightarrow{b} 3 \xrightarrow{c} 4$ , with  $|a| = |b| = |c| = -1$  and  $d_A = 0$ . Then  $M = 4$ . Choose  $(\sigma(1), \sigma(2), \sigma(3), \sigma(4)) = (2, 4, 1, 3)$ , and put  $U = P_1/(abA)$ . The standard objects and their kernels are

| $v$ | $\Delta_v$ | $K_v$                   | $\Delta^\sigma = (S_2, S_4, U, P_3).$ |
|-----|------------|-------------------------|---------------------------------------|
| 1   | $U$        | $abA \cong \Delta_3[2]$ |                                       |
| 2   | $S_2$      | $bA \cong \Delta_3[1]$  |                                       |
| 3   | $P_3$      | 0                       |                                       |
| 4   | $S_4$      | 0                       |                                       |

(3.6)

The object  $P_3$  has cohomology  $S_3$  in degree 0 and  $S_4$  in degree  $-1$ . Thus the two non-zero kernels in (3.6) occupy degrees  $[-3, -2]$  and  $[-2, -1]$ , respectively, and both are single allowed factors in  $\mathcal{H}_A^4$ .

**Example 3.3** (A graded quiver of type  $D_4$ ). Take  $Q : 1 \xrightarrow{a} 2 \xrightarrow{b} 3, 2 \xrightarrow{c} 4$ , with  $|a| = |c| = -1$ ,  $|b| = 0$ , and  $d_A = 0$ . Then  $M = 3$ . Choose  $(\sigma(1), \sigma(2), \sigma(3), \sigma(4)) = (1, 3, 2, 4)$ , and put  $V = P_2/cA$ . Then

$$\begin{array}{c|c|c} v & \Delta_v & K_v \\ \hline 1 & S_1 & aA \cong P_2[1] \\ 2 & V & cA \cong \Delta_4[1] \\ 3 & S_3 & 0 \\ 4 & S_4 & 0 \end{array} \quad \Delta^\sigma = (S_1, S_3, V, S_4). \quad (3.7)$$

The kernel  $K_1$  has the two-factor filtration

$$0 \subset acA \subset aA, \quad acA \cong \Delta_4[2], \quad aA/acA \cong \Delta_2[1]. \quad (3.8)$$

Both factors belong to  $\mathcal{H}_A^3$ , with  $1 <_\sigma 4$  and  $1 <_\sigma 2$ . The filtration (3.8) refines the shifted-projective description of  $K_1$  into shifted standard factors.

**Example 3.4** (A differential and the required width). Take arrows  $a : 1 \rightarrow 2$ ,  $b : 2 \rightarrow 3$ , and  $c : 1 \rightarrow 3$ , with

$$|a| = |b| = 0, \quad |c| = -1, \quad d_A(c) = ab, \quad d_A(a) = d_A(b) = 0. \quad (3.9)$$

Here  $M = 2$ , but  $H^q(A) = 0$  for  $q \neq 0$  and  $B = H^0(A) = \mathbb{k}(1 \xrightarrow{a} 2 \xrightarrow{b} 3)/(ab)$ . For  $(\sigma(1), \sigma(2), \sigma(3)) = (1, 3, 2)$ ,

$$(\Delta_1, \Delta_2, \Delta_3) = (S_1, P_2, S_3), \quad K_1 = \mathbb{k}\{a, ab, c\}, \quad K_2 = K_3 = 0. \quad (3.10)$$

There is a filtration by dg submodules

$$0 \subset aA \subset K_1, \quad aA \cong \Delta_2, \quad K_1/aA \cong \Delta_3[1]. \quad (3.11)$$

Theorem 3.1 gives a structure of width 2. Although  $K_1 \simeq S_2$  and all the objects in (3.10) lie in  $\mathcal{H}_A^1$ , the factor  $\Delta_3[1]$  does not. There is no width-1 structure for this ordering: Proposition 2.4 fixes the standard objects. Write  $[X]$  for the class of a module  $X$  in  $K_0(\text{mod } B)$ , whose basis is  $[S_1], [S_2], [S_3]$ . A filtration of  $S_2$  by  $P_2$  and  $S_3$  in  $\text{mod } B$  would require

$$[S_2] = x[P_2] + y[S_3] = x[S_2] + (x+y)[S_3], \quad x, y \in \mathbb{Z}_{\geq 0},$$

which forces  $x = 1$  and  $y = -1$ , a contradiction.

## 4 Filtrations, width, and opposite algebras

Fix a structure satisfying Definition 1.1. For an integer  $\ell \geq 1$ , define

$$I_w^\ell = \{s \in \mathbb{Z} : 0 \leq s < \ell, \Delta_w[s] \in \mathcal{H}_A^\ell\}, \quad \mathcal{S}_{>v}^\ell = \{\Delta_w[s] : v <_\sigma w, s \in I_w^\ell\}. \quad (4.1)$$

For  $1 \leq v, w \leq n$  and  $t \in \mathbb{Z}$ , put

$$b_{vw,t} = \dim_{\mathbb{k}} \text{Hom}_{\mathcal{D}_{\text{fd}}(A)}(P_v, \nabla_w[t]) = \dim_{\mathbb{k}} H^t(\nabla_w)e_v. \quad (4.2)$$

**Proposition 4.1** (A criterion for shifted filtrations). *For  $X \in \mathcal{Q}_v$ , where  $\mathcal{Q}_v$  is (2.6),*

$$X \in \text{Filt}_{\mathcal{H}_A^\ell}(\mathcal{S}_{>v}^\ell) \iff \text{Hom}_{\mathcal{D}_{\text{fd}}(A)}(X, \nabla_w[t]) = 0 \quad (4.3)$$

*for  $v <_\sigma w$ ,  $t \notin I_w^\ell$ .*

When (4.3) holds,  $X$  has a filtration by decreasing weights whose weight- $w$  factor is

$$G_w(X) \cong \bigoplus_{s \in I_w^\ell} \Delta_w[s]^{\oplus \dim_{\mathbb{k}} \text{Hom}_{\mathcal{D}_{\text{fd}}(A)}(X, \nabla_w[s])}. \quad (4.4)$$

The category  $\text{Filt}_{\mathcal{H}_A^\ell}(\mathcal{S}_{>v}^\ell)$  is closed under direct summands. For  $X = K_v$ , the multiplicities in (4.4) are  $b_{vw,s}$ .

*Proof.* We prove necessity, construct the filtration for sufficiency, and then deduce the assertions about multiplicities and direct summands.

Necessity. By (2.5), an allowed factor  $\Delta_u[s]$  has a non-zero morphism to  $\nabla_w[t]$  only when  $u = w$  and  $t = s$ . The long exact sequences of a filtration give the vanishings in (4.3).

Sufficiency. Put  $a = \sigma^{-1}(v)$  and apply the evaluation construction (2.2) to  $X$  in the exceptional subcategory  $\mathcal{Q}_v$ . The resulting triangles, combined by the octahedral axiom, give an ordered tower with successive factors in  $\text{thick}_{\mathcal{D}_{\text{fd}}(A)}(\Delta_{\sigma(b)})$  for  $b = n, n-1, \dots, a+1$ . If  $a = n$ , then  $\mathcal{Q}_v = 0$  and the tower is empty. For each weight  $w$ , exceptionality identifies  $\text{thick}_{\mathcal{D}_{\text{fd}}(A)}(\Delta_w)$  with  $\mathcal{D}^b(\text{mod } \mathbb{k})$ , so its factor is a finite direct sum of shifts of  $\Delta_w$ . Applying  $\text{Hom}_{\mathcal{D}_{\text{fd}}(A)}(-, \nabla_w[t])$  to the tower kills every factor of weight different from  $w$ . The pairing (2.3) therefore identifies the multiplicity of  $\Delta_w[t]$  with  $\dim_{\mathbb{k}} \text{Hom}_{\mathcal{D}_{\text{fd}}(A)}(X, \nabla_w[t])$ . The vanishings in (4.3) give precisely (4.4). Every factor and, by extension closure, every intermediate object belongs to  $\mathcal{H}_A^\ell$ . Splitting the finite direct sums refines the tower to a filtration by  $\mathcal{S}_{>v}^\ell$ .

Both  $\mathcal{Q}_v$  and the vanishing conditions in (4.3) are closed under direct summands. For  $v <_\sigma w$ , applying  $\text{Hom}_{\mathcal{D}_{\text{fd}}(A)}(-, \nabla_w[t])$  to (1.10) gives  $\text{Hom}_{\mathcal{D}_{\text{fd}}(A)}(K_v, \nabla_w[t]) \cong \text{Hom}_{\mathcal{D}_{\text{fd}}(A)}(P_v, \nabla_w[t])$ . This proves the final assertion using (4.2).  $\square$

Arbitrary filtrations still need not have invariant multiplicities. For  $v <_\sigma w$  and  $0 \leq s < m-1$  with  $\Delta_w[s], \Delta_w[s+1] \in \mathcal{H}_A^m$ , the conflations

$$0 \longrightarrow \Delta_w[s] \xrightarrow{1} \Delta_w[s] \dashrightarrow, \quad \Delta_w[s] \longrightarrow 0 \longrightarrow \Delta_w[s+1] \dashrightarrow \quad (4.5)$$

give a two-factor filtration of zero. In contrast, the multiplicities in the ordered factors (4.4) are fixed by the left-dual pairing. If  $c_{vw,s}$  counts factors  $\Delta_w[s]$  in any filtration of  $K_v$ , Euler additivity gives

$$\sum_{s=0}^{m-1} (-1)^s c_{vw,s} = \sum_{s=0}^{m-1} (-1)^s b_{vw,s} \quad (v <_\sigma w). \quad (4.6)$$

Proposition 4.1 supplies a filtration with  $c_{vw,s} = b_{vw,s}$  for all  $v <_\sigma w$  and  $0 \leq s < m$ .

**Corollary 4.2** (Least width). *Suppose  $(A, \sigma)$  admits a structure of Definition 1.1 for at least one width. For its canonical standard objects and specified left-dual objects, put*

$$\begin{aligned} \alpha_A &= -\min\{q : H^q(A) \neq 0\}, \\ \alpha_w &= -\min\{q : H^q(\Delta_w) \neq 0\}, \quad \beta_w = \max\{q : H^q(\nabla_w) \neq 0\}. \end{aligned} \quad (4.7)$$

Then  $\alpha_A, \alpha_w, \beta_w \geq 0$ , and the least width is

$$m_{\min}(A, \sigma) = 1 + \max \left\{ \alpha_A, \max_{1 \leq w \leq n} (\alpha_w + \beta_w) \right\}. \quad (4.8)$$

*Proof.* Proposition 2.4 makes  $\Delta_w$  and  $\nabla_w$  independent of the chosen width. Its proof, together with (2.3), also gives

$$H^t(\nabla_w)e_v = 0 \quad (w <_\sigma v), \quad H^t(\nabla_w)e_w \cong \begin{cases} \mathbb{k}, & t = 0, \\ 0, & t \neq 0. \end{cases} \quad (4.9)$$

Thus  $\beta_w \geq 0$ , and if  $\beta_w > 0$ , then  $b_{vw, \beta_w} \neq 0$  for some  $v <_\sigma w$ . Since  $H^0(\Delta_w) \neq 0$ , membership of a shifted standard is equivalent to

$$\Delta_w[s] \in \mathcal{H}_A^\ell \iff 0 \leq s \leq \ell - 1 - \alpha_w \quad (s \geq 0).$$

Necessity in (4.3) therefore forces  $\alpha_w + \beta_w \leq \ell - 1$  whenever width  $\ell$  is possible; when  $\beta_w = 0$ , the same inequality follows from  $\Delta_w \in \mathcal{H}_A^\ell$ . Also  $\alpha_A \leq \ell - 1$  is necessary.

Conversely, these inequalities put every non-zero factor  $\Delta_w[s]$  of the ordered kernel tower in  $\mathcal{H}_A^\ell$ , with  $0 \leq s < \ell$ . Proposition 4.1 gives  $K_v \in \text{Filt}_{\mathcal{H}_A^\ell}(\mathcal{S}_{>v}^\ell)$ . The objects  $P_v$  and  $\Delta_v$  also lie in  $\mathcal{H}_A^\ell$ , so their presentation triangles are conflations. This proves (4.8).  $\square$



**Proposition 4.3** (Injective presentations and opposite algebras). *Set  $\widehat{\nabla}_v = \nabla_v[m-1]$  and  $I_v = \mathbb{D}(Ae_v)[m-1]$ . There are conflations*

$$\begin{aligned} \widehat{\nabla}_v &\longrightarrow I_v \longrightarrow L_v \dashrightarrow \quad \text{in } \mathcal{H}_A^m, \\ L_v &\in \text{Filt}_{\mathcal{H}_A^m} \{ \widehat{\nabla}_w[-s] : v <_\sigma w, 0 \leq s < m, \widehat{\nabla}_w[-s] \in \mathcal{H}_A^m \}. \end{aligned} \quad (4.10)$$

The algebra  $A^{\text{op}}$  satisfies Definition 1.1 for the same  $m$  and  $\sigma$ , with standards  $\mathbb{D}\nabla_v$  and specified left-dual objects  $\mathbb{D}\Delta_v$ .

*Proof.* Decompose  $I_v$  along the full exceptional sequence  $(\nabla_{\sigma(n)}, \dots, \nabla_{\sigma(1)})$ . The factors of  $I_v$  occur in increasing order in  $\leq_\sigma$ . The pairing (2.3) and the identity

$$\text{Hom}_{\mathcal{D}_{\text{fd}}(A)}(\Delta_w, I_v[t]) \cong \mathbb{D}(H^{1-m-t}(\Delta_w)e_v)$$

identify the weight- $w$  factor as

$$\bigoplus_{s \geq 0} \nabla_w[m-1-s]^{\oplus \dim_{\mathbb{k}} H^{-s}(\Delta_w)e_v}. \quad (4.11)$$

By (2.8), weights  $w <_\sigma v$  contribute zero, and the first non-zero factor is  $\widehat{\nabla}_v$ . Every occurring  $s$  satisfies  $s \leq \alpha_w$ ; hence  $\beta_w + s \leq m-1$  by Corollary 4.2. Thus all factors in (4.11) belong to  $\mathcal{H}_A^m$ , giving (4.10).

The contravariant equivalence  $\mathbb{D}(-)[m-1] : \mathcal{H}_A^m \rightarrow \mathcal{H}_{A^{\text{op}}}^m$  takes (4.10) to the required projective presentation:

$$\mathbb{D}L_v[m-1] \longrightarrow Ae_v \longrightarrow \mathbb{D}\nabla_v \dashrightarrow .$$

Its kernel factors are  $(\mathbb{D}\nabla_w)[s]$  with  $v <_\sigma w$ . Vector-space duality reverses the costandard exceptional sequence, giving  $(\mathbb{D}\nabla_{\sigma(1)}, \dots, \mathbb{D}\nabla_{\sigma(n)})$ . Finally,

$$\text{Hom}_{\mathcal{D}_{\text{fd}}(A^{\text{op}})}(\mathbb{D}\nabla_u, \mathbb{D}\Delta_v[t]) \cong \text{Hom}_{\mathcal{D}_{\text{fd}}(A)}(\Delta_v, \nabla_u[t])$$

proves the asserted left-dual pairing.  $\square$

## 5 Framework for replacing (Q1)

In Sections 6–12, retain the standing assumptions (1.1), the order (1.8), and condition Q2. The objects  $\Delta_v \in \mathcal{H}_A^m$  are not assumed exceptional. Each proposal retains the *brick condition*

$$\text{End}_{\mathcal{H}_A^m}(\Delta_v) \cong \mathbb{k} \quad (1 \leq v \leq n), \quad (5.1)$$

unless (5.1) is explicitly proved from stronger conditions. The purpose of replacing condition Q1 is to obtain homological smoothness as a theorem. Smoothness as an additional assumption is considered only for comparison in Section 9.

For a finite-dimensional  $B$ -module  $M$ , write  $[M : S_w]$  for its composition multiplicity. Since  $B$  is split basic,

$$[M : S_w] = \dim_{\mathbb{k}} Me_w, \quad \text{Hom}_{\mathcal{D}_{\text{fd}}(A)}(P_w, X[t]) \cong H^t(X)e_w \quad (X \in \mathcal{D}_{\text{fd}}(A), t \in \mathbb{Z}). \quad (5.2)$$

The equality  $\text{Hom}_{\mathcal{H}_A^m}(X, Y) = \text{Hom}_{\mathcal{D}_{\text{fd}}(A)}(X, Y)$  applies only when  $X, Y \in \mathcal{H}_A^m$ .

**Proposition 5.1** (Generation from the filtration axiom). *Condition Q2 alone implies*

$$\Delta_v \in \text{per } A \quad (1 \leq v \leq n), \quad \text{thick}_{\mathcal{D}_{\text{fd}}(A)}(\Delta_1, \dots, \Delta_n) = \text{per } A. \quad (5.3)$$

*Proof.* The greatest weight has  $K_{\sigma(n)} = 0$  and  $\Delta_{\sigma(n)} \cong P_{\sigma(n)}$ . Descending induction using (1.11) and (1.10) makes every  $K_v$  and  $\Delta_v$  perfect. Conversely, the same triangles put every  $P_v$  in the thick subcategory generated by the standards.  $\square$



**Proposition 5.2** (The classical projective test). *For  $m = 1$ , the brick condition (5.1), condition Q2, and either of the conditions*

$$\mathrm{Hom}_B(P_w, \Delta_v) = 0 \quad (v <_\sigma w), \quad (5.4)$$

$$\mathrm{Hom}_B(\Delta_w, \Delta_v) = 0 \quad (v <_\sigma w) \quad (5.5)$$

*are equivalent to classical quasi-heredity. Here the equivalence  $\mathcal{H}_A^1 \simeq \mathrm{mod} B$  identifies  $P_w$  with  $e_w B$ .*

*Proof.* First prove the equivalence of (5.4) and (5.5).

Projective test  $\Rightarrow$  standard test. Precomposition with the epimorphism  $P_w \twoheadrightarrow \Delta_w$  embeds  $\mathrm{Hom}_B(\Delta_w, \Delta_v)$  in  $\mathrm{Hom}_B(P_w, \Delta_v)$ .

Standard test  $\Rightarrow$  projective test. The filtration of  $K_w$  and (5.5) give  $\mathrm{Hom}_B(K_w, \Delta_v) = 0$  for  $v <_\sigma w$ . Apply  $\mathrm{Hom}_B(-, \Delta_v)$  to  $0 \rightarrow K_w \rightarrow P_w \rightarrow \Delta_w \rightarrow 0$ .

Under either test,  $\Delta_v$  is supported on weights at most  $v$ , and  $\mathrm{Hom}_B(K_v, \Delta_v) = 0$ . Consequently

$$\mathbb{k} \cong \mathrm{End}_B(\Delta_v) \cong \mathrm{Hom}_B(P_v, \Delta_v) \cong \Delta_v e_v.$$

The quotient  $P_v \rightarrow \Delta_v$  has simple top  $S_v$ , with  $S_v$  occurring once. The projective kernels are filtered by standards of larger weights. These are the classical axioms [BB26, §2.1]; classical standards conversely satisfy both tests.  $\square$

The brick condition cannot be omitted from Proposition 5.2. The ordinary algebra  $B = \mathbb{k}[x]/(x^2)$  with one standard  $\Delta_1 = P_1 = B$  and  $K_1 = 0$  satisfies both off-diagonal tests, whose index sets are empty, but is not quasi-hereditary.

**Proposition 5.3** (Cohomological support). *Assume condition Q2 and*

$$H^t(\Delta_v)e_w = 0 \quad (v <_\sigma w, t \in \mathbb{Z}). \quad (5.6)$$

*Then*

$$\mathrm{Hom}_{\mathcal{D}_{\mathrm{fd}}(A)}(\Delta_w, \Delta_v[t]) = 0 \quad (v <_\sigma w, t \in \mathbb{Z}), \quad (5.7)$$

$$\mathrm{Hom}_{\mathcal{D}_{\mathrm{fd}}(A)}(\Delta_v, \Delta_v[t]) \cong H^t(\Delta_v)e_v \quad (1 \leq v \leq n, t \in \mathbb{Z}). \quad (5.8)$$

*In particular, (5.1) is equivalent to  $[H^0(\Delta_v) : S_v] = 1$  for every  $v$  under these hypotheses.*

*Proof.* Fix  $v$  and descend through the weights  $w >_\sigma v$ . Equation (5.2) gives

$$\mathrm{Hom}_{\mathcal{D}_{\mathrm{fd}}(A)}(P_w, \Delta_v[t]) = 0 \quad (t \in \mathbb{Z}).$$

Once the standards of weight greater than  $w$  are orthogonal to  $\Delta_v$  in every degree, their shifted filtration of  $K_w$  gives

$$\mathrm{Hom}_{\mathcal{D}_{\mathrm{fd}}(A)}(K_w, \Delta_v[t]) = 0 \quad (t \in \mathbb{Z}).$$

The presentation of  $\Delta_w$  proves (5.7). The filtration of  $K_v$  now gives the same vanishing for  $K_v$ ; precomposition with  $p_v : P_v \rightarrow \Delta_v$  proves (5.8).  $\square$

**Example 5.4** (A non-smooth dga with one brick standard). Let

$$R = \mathbb{k}[\varepsilon]/(\varepsilon^2), \quad |\varepsilon| = -1, \quad d_R = 0, \quad m = 2, \quad \Delta_1 = P_1 = R, \quad K_1 = 0. \quad (5.9)$$

The object  $R \in \mathcal{H}_R^2$  is a brick, and

$$\mathrm{Hom}_{\mathcal{D}_{\mathrm{fd}}(R)}(R, R[t]) \cong \begin{cases} \mathbb{k}, & t = 0, -1, \\ 0, & \text{otherwise.} \end{cases} \quad (5.10)$$

To verify non-smoothness, put  $S = R/(\varepsilon)$ . A semifree resolution  $F \rightarrow S$  is given by

$$F = \bigoplus_{a \geq 0} x_a R, \quad |x_a| = -2a, \quad d_F(x_0) = 0, \quad d_F(x_a) = x_{a-1}\varepsilon \quad (a \geq 1), \quad (5.11)$$

where the augmentation sends  $x_0$  to 1 and  $x_a$  to 0 for  $a > 0$ . Each  $x_a\varepsilon$  is the boundary of  $x_{a+1}$ , so  $H^*(F) = S$ . Applying the dg Hom complex into  $S$  gives

$$\mathrm{Hom}_{\mathcal{D}_{\mathrm{fd}}(R)}(S, S[t]) \cong \begin{cases} \mathbb{k}, & t \in 2\mathbb{Z}_{\geq 0}, \\ 0, & \text{otherwise.} \end{cases} \quad (5.12)$$

Thus  $S \notin \mathrm{per} R$ , since a perfect object has finite-dimensional total derived Hom into  $S$ . Smoothness would put every cohomologically finite-dimensional module in  $\mathrm{per} R$ : tensor a finite construction of the diagonal bimodule with that module. Therefore  $R$  is not smooth and  $\mathrm{per} R \subsetneq \mathcal{D}_{\mathrm{fd}}(R)$ .

## 6 Bricks and backward Hom-orthogonality

Replace condition **Q1** by (5.1) and

$$\mathrm{Hom}_{\mathcal{H}_A^m}(\Delta_w, \Delta_v) = 0 \quad (v <_\sigma w). \quad (\mathrm{H})$$

Together with condition **Q2**, (H) has the correct classical specialisation by Proposition 5.2. For  $m > 1$ , a factor  $\Delta_u[s]$  of  $K_w$  contributes  $\mathrm{Hom}_{\mathcal{D}_{\mathrm{fd}}(A)}(\Delta_u, \Delta_v[-s])$  to  $\mathrm{Hom}_{\mathcal{D}_{\mathrm{fd}}(A)}(K_w, \Delta_v)$ . Equation (H) controls only  $s = 0$ .

**Example 6.1** (Backward extensions on a graded fork). Take

$$A = \mathbb{k}(1 \xrightarrow{a} 3 \xleftarrow{b} 2), \quad |a| = |b| = -1, \quad d_A = 0, \quad m = 2,$$

with  $e_1a = a = ae_3$ ,  $e_2b = b = be_3$ , and order  $1 < 2 < 3$ . Choose

$$(\Delta_1, \Delta_2, \Delta_3) = (P_1, S_2, P_3), \quad (K_1, K_2, K_3) = (0, P_3[1], 0). \quad (6.1)$$

The short exact sequence of dg modules

$$0 \longrightarrow P_3[1] \xrightarrow{b} P_2 \longrightarrow S_2 \longrightarrow 0 \quad (6.2)$$

induces the required conflation in  $\mathcal{H}_A^2$ . Every  $\Delta_v$  is individually exceptional and  $H^0(\Delta_v) = S_v$ . Nevertheless,

$$\mathrm{Hom}_{\mathcal{D}_{\mathrm{fd}}(A)}(\Delta_w, \Delta_v[t]) \cong \begin{cases} \mathbb{k}, & (w, v, t) = (3, 1, -1) \text{ or } (2, 1, 1), \\ 0, & w > v \text{ otherwise.} \end{cases} \quad (6.3)$$

Indeed,  $\mathrm{Hom}_{\mathcal{D}_{\mathrm{fd}}(A)}(P_2, P_1[t]) = 0$  for every  $t$ ; applying derived Hom into  $P_1$  to (6.2) computes the pair  $(2, 1)$ . Equation (6.3) proves (H) but disproves backward orthogonality in degrees 1 and  $-1$ . The algebra is smooth; the failure concerns the proposed standards and their order.

**Example 6.2** (Individual exceptionality does not ensure smoothness). Let  $A$  have vertices 1, 2, arrows  $a : 1 \rightarrow 2$  and  $b : 2 \rightarrow 1$ , with  $|a| = |b| = -1$ ,  $d_A = 0$ , and relations  $ab = ba = 0$ . Paths are concatenated from left to right. Take  $m = 2$ , order  $1 < 2$ , and  $\Delta_i = P_i$ ,  $K_i = 0$  for  $i = 1, 2$ . Since  $e_i A e_i = \mathbb{k}$ , both standards are exceptional, and (H) holds. But

$$\mathrm{Hom}_{\mathcal{D}_{\mathrm{fd}}(A)}(P_2, P_1[-1]) = \mathbb{k}a.$$

An alternating semifree resolution  $F \rightarrow S_1$  has generators  $x_j$  of degree  $-2j$ , at vertex 1 for even  $j$  and at vertex 2 for odd  $j$ . Its differential is  $d_F(x_j) = x_{j-1}a$  for odd  $j$  and  $d_F(x_j) = x_{j-1}b$

for even  $j > 0$ ;  $d_F(x_0) = 0$ . The augmentation  $F \rightarrow S_1$  sends  $x_0$  to a generator of  $S_1$  and every  $x_j$ ,  $j > 0$ , to zero. The relations  $ab = ba = 0$  make  $d_F^2 = 0$ , and the augmentation is a quasi-isomorphism. Hence

$$\mathrm{Hom}_{\mathcal{D}_{\mathrm{fd}}(A)}(S_1, S_1[4j]) \cong \mathbb{k} \quad (j \geq 0). \quad (6.4)$$

The module  $S_1$  is not perfect, so  $A$  is not smooth. Thus even supplementing (H) with individual exceptionality leaves a non-smooth example.

| Property under (H)                                | Result with (5.1) and condition Q2            |
|---------------------------------------------------|-----------------------------------------------|
| Classical specialisation                          | Yes; Proposition 5.2.                         |
| Bricks; backward degree-zero Hom                  | Required.                                     |
| Standards generate per $A$                        | Yes; Proposition 5.1.                         |
| Backward Hom in every degree                      | No; Example 6.1.                              |
| Self-exceptionality                               | Not forced; Example 5.4.                      |
| Standards generate $\mathcal{D}_{\mathrm{fd}}(A)$ | Not forced; Example 5.4.                      |
| Smoothness as a consequence                       | No, even with exceptional terms; Example 6.2. |

## 7 Cohomological support with a simple diagonal

Replace condition Q1 by

$$\begin{aligned} H^q(\Delta_v)e_w &= 0 \quad (v <_\sigma w, \quad q \in \mathbb{Z}), \\ H^q(\Delta_v)e_v &\cong \begin{cases} \mathbb{k}, & q = 0, \\ 0, & q \neq 0 \end{cases} \quad (1 \leq v \leq n). \end{aligned} \quad (\mathrm{C})$$

Equivalently,  $H^0(\Delta_v)$  has weights at most  $v$ , with  $S_v$  occurring once, and  $H^q(\Delta_v)$  has weights strictly smaller than  $v$  for  $q < 0$ . No Hom- or Ext-vanishing between standards is imposed. In the formulation (C) and condition Q2, neither brickness, exceptionality, fullness, nor smoothness is an additional axiom.

**Proposition 7.1** (An equivalent composition-factor definition). *Under condition Q2, (C) is equivalent to condition Q1. In particular, the brick condition, fullness in  $\mathcal{D}_{\mathrm{fd}}(A) = \mathrm{per} A$ , and smoothness follow.*

*Proof.* (C)  $\Rightarrow$  condition Q1. Proposition 5.3 proves backward orthogonality in every degree. Its diagonal identity (5.8) and (C) prove exceptionality of each  $\Delta_v$ .

Condition Q1  $\Rightarrow$  (C). Apply Proposition 2.4, specifically (2.8). Proposition 2.1 and Corollary 2.2 give the remaining assertions.  $\square$

**Proposition 7.2** (Smoothness directly from composition factors). *Under the standing assumptions (1.1), (C) and condition Q2 imply*

$$S_v \in \mathrm{per} A \quad (1 \leq v \leq n), \quad \mathcal{D}_{\mathrm{fd}}(A) = \mathrm{per} A, \quad A \in \mathrm{per}(A^{\mathrm{op}} \otimes_{\mathbb{k}} A). \quad (7.1)$$

*Proof. Step 1: descending induction makes the standards perfect.* Condition Q2 gives  $\Delta_{\sigma(n)} \cong P_{\sigma(n)}$ . For each  $v$ , once the standards  $\Delta_w$  with  $w >_\sigma v$  are perfect, their shifted filtration (1.11) makes  $K_v$  perfect. The triangle (1.10) then makes  $\Delta_v$  perfect.

*Step 2: ascending induction makes the simples perfect.* Since  $H^1(K_v) = 0$ , the morphism  $H^0(p_v) : e_v B \rightarrow H^0(\Delta_v)$  is surjective. Condition (C) gives  $[H^0(\Delta_v) : S_v] = 1$ , so  $H^0(\Delta_v)$  has simple top  $S_v$ . Let  $q_v : H^0(\Delta_v) \rightarrow S_v$  be a top quotient, let  $a_v : \Delta_v \rightarrow S_v$  be the composite of the canonical truncation morphism  $\Delta_v \rightarrow H^0(\Delta_v)$  with  $q_v$ , and take the triangle

$$T_v \longrightarrow \Delta_v \xrightarrow{a_v} S_v \longrightarrow T_v[1]. \quad (7.2)$$

The cohomology of  $T_v$  satisfies

$$H^q(T_v) \cong \begin{cases} H^q(\Delta_v), & q < 0, \\ \text{Ker } q_v, & q = 0, \\ 0, & q > 0. \end{cases} \quad (7.3)$$

By (C), every composition factor of every  $H^q(T_v)$  is  $S_w$  with  $w <_\sigma v$ . Finite truncation triangles and composition series therefore give  $T_v \in \text{thick}_{\mathcal{D}_{\text{fd}}(A)}\{S_w : w <_\sigma v\}$ . For the least weight,  $T_{\sigma(1)} = 0$  and  $S_{\sigma(1)} \cong \Delta_{\sigma(1)}$  is perfect. Ascending induction and (7.2) make every  $S_v$  perfect. Every object of  $\mathcal{D}_{\text{fd}}(A)$  is obtained from the  $S_v$  by finite truncation triangles and composition series, so  $\mathcal{D}_{\text{fd}}(A) = \text{per } A$ .

*Step 3: perfect simples imply smoothness.* The radical quotient  $B/\text{rad } B \cong \mathbb{k}^n$  is separable over  $\mathbb{k}$  and perfect over  $A$ . The criterion [RS22, Theorem 5.8] gives  $A \in \text{per}(A^{\text{op}} \otimes_{\mathbb{k}} A)$ .  $\square$

The obstruction excluded by (C) is explicit: for  $R = \mathbb{k}[\varepsilon]/(\varepsilon^2)$ ,  $|\varepsilon| = -1$ ,  $d_R = 0$ , and  $\Delta_1 = R$ , the group  $H^{-1}(\Delta_1)e_1 = \mathbb{k}\varepsilon$  violates the diagonal condition. Example 5.4 proves that removing precisely that restriction admits a non-smooth algebra. There is no admitted non-smooth counterexample under (C) and condition Q2, by Proposition 7.1.

Negative cohomology of standards is still permitted at smaller weights. For example, take  $A = \mathbb{k}(1 \xrightarrow{a} 2)$  with  $|a| = -1$ ,  $d_A = 0$ ,  $m = 2$ , and order  $2 <_\sigma 1$ . Then  $\Delta_1 = P_1$ ,  $\Delta_2 = P_2$ ,  $K_1 = K_2 = 0$ , and  $H^{-1}(\Delta_1) = S_2$ .

| Property under (C)                                             | Result with condition Q2                                                                                      |
|----------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------|
| Classical specialisation                                       | Yes; Corollary 2.5.                                                                                           |
| Bricks and backward Hom in every degree                        | Consequences of Proposition 5.3.                                                                              |
| Exceptional standard sequence                                  | A consequence, not an axiom.                                                                                  |
| Generation of $\text{per } A$ and $\mathcal{D}_{\text{fd}}(A)$ | Both hold.                                                                                                    |
| Smoothness as a consequence                                    | Yes; the direct proof is Proposition 7.2.                                                                     |
| Negative cohomology of standards                               | Allowed at strictly smaller weights.                                                                          |
| Non-smooth algebras excluded                                   | Yes, as a consequence of the axioms. In particular, the dg dual numbers fail the diagonal requirement in (C). |
| Additional restriction compared with (Q1) and (Q2)             | None; Proposition 7.1 proves equivalence.                                                                     |

## 8 Allowing the diagonal in negative cohomology

Retain (5.1) and replace condition Q1 by

$$H^q(\Delta_v)e_w = 0 \quad (v <_\sigma w, q \in \mathbb{Z}). \quad (C')$$

Under condition Q2, Proposition 5.3 shows that  $[H^0(\Delta_v) : S_v] = 1$  follows from brickness. It also proves all backward derived orthogonality and

$$\text{Hom}_{\mathcal{D}_{\text{fd}}(A)}(\Delta_v, \Delta_v[t]) \cong H^t(\Delta_v)e_v = 0 \quad (t > 0). \quad (8.1)$$

Negative self-Hom spaces remain unrestricted by (C').

The data

$$A = \mathbb{k}[\varepsilon]/(\varepsilon^2), \quad |\varepsilon| = -1, \quad d_A = 0, \quad m = 2, \quad \Delta_1 = P_1 = A, \quad K_1 = 0$$

satisfy (5.1), (C'), and condition Q2. Example 5.4 proves that  $A$  is not smooth and that  $\text{thick}_{\mathcal{D}_{\text{fd}}(A)}(\Delta_1) = \text{per } A \subsetneq \mathcal{D}_{\text{fd}}(A)$ . Even the pairing object cannot be a left-dual exceptional object: requiring  $\text{Hom}_{\mathcal{D}_{\text{fd}}(A)}(A, N[t]) = \mathbb{k}$  for  $t = 0$  and zero otherwise forces  $N \simeq \mathbb{k}$ , which is neither perfect nor exceptional by (5.12).

**Proposition 8.1** (Smoothness and the diagonal). *Assume (5.1), (C'), and condition Q2. Then the following are equivalent:*

- (i)  $A$  is homologically smooth;
- (ii)  $H^q(\Delta_v)e_v = 0$  for all  $v$  and all  $q < 0$ ;
- (iii)  $\Delta^\sigma$  is an exceptional sequence.

*Proof.* We prove (i)  $\Rightarrow$  (ii)  $\Rightarrow$  (iii)  $\Rightarrow$  (i).

(i)  $\Rightarrow$  (ii). Put  $E_v = \mathbf{R}\mathrm{End}_A(\Delta_v)$ . Equations (5.8) and (5.1) imply that  $E_v$  is proper and connective, with  $H^0(E_v) \cong \mathbb{k}$ . Choose homotopically projective representatives of the standards. In their dg endomorphism category, the backward morphism complexes are acyclic by (5.7). Replacing those complexes by zero gives a quasi-equivalent upper triangular dg category, whose diagonal dg algebras are the  $E_v$ . Proposition 5.1 makes that dg category Morita equivalent to  $A$ . The smoothness theorem for dg gluings [Orl16, Theorem 3.25] implies that every  $E_v$  is smooth. Now [RS22, Proposition 7.4] gives  $E_v \simeq \mathbb{k}$ . Equation (5.8) proves (ii).

(ii)  $\Rightarrow$  (iii). The brick condition and (5.8) give the degree-zero diagonal multiplicity. Thus (ii) strengthens (C') to (C). Apply Proposition 7.1.

(iii)  $\Rightarrow$  (i). Apply Corollary 2.2. □

| Property under (C')                                               | Result with (5.1) and condition Q2             |
|-------------------------------------------------------------------|------------------------------------------------|
| Classical specialisation                                          | Yes; the negative degrees disappear.           |
| Backward Hom in every degree                                      | Yes; (5.7).                                    |
| Bricks; positive self-Hom vanishing                               | Yes; (8.1).                                    |
| Standards generate per $A$                                        | Yes; (5.3).                                    |
| Self-exceptionality; generation of $\mathcal{D}_{\mathrm{fd}}(A)$ | Not forced; Example 5.4.                       |
| Smoothness as a consequence                                       | No; the dg dual numbers are admitted.          |
| Exact obstruction to smoothness                                   | Negative diagonal cohomology; Proposition 8.1. |

## 9 Adding smoothness as an assumption

For comparison, replace condition Q1 by (5.1) and

$$(C') \text{ holds, and } A \in \mathrm{per}(A^{\mathrm{op}} \otimes_{\mathbb{k}} A). \quad (C_{\mathrm{sm}})$$

Proposition 8.1 proves that  $(C_{\mathrm{sm}})$ , together with condition Q2, is equivalent to Definition 1.1. In particular, the attempted allowance of negative diagonal cohomology disappears.

The algebra  $\mathbb{k}[\varepsilon]/(\varepsilon^2)$  with  $|\varepsilon| = -1$  and  $d = 0$  is excluded precisely by the smoothness assumption; it satisfies the other requirements by Example 5.4. No non-smooth example satisfies  $(C_{\mathrm{sm}})$ , by definition. The defect of this proposal is axiomatic: smoothness is required rather than obtained from the standard-module conditions.

| Property under $(C_{\mathrm{sm}})$                            | Result with (5.1) and condition Q2                                  |
|---------------------------------------------------------------|---------------------------------------------------------------------|
| Classical specialisation                                      | Yes.                                                                |
| Backward orthogonality and self-exceptionality                | Consequences of Proposition 8.1.                                    |
| Generation of $\mathrm{per} A = \mathcal{D}_{\mathrm{fd}}(A)$ | Yes.                                                                |
| Negative diagonal cohomology                                  | Necessarily zero.                                                   |
| Non-smooth examples                                           | Excluded by assumption.                                             |
| Smoothness as a consequence of standard-module axioms         | No; smoothness is a separate hypothesis.                            |
| Distinct class from (C)                                       | No; the two formulations are equivalent under the filtration axiom. |

## 10 Projective tests with non-negative shifts

Retain the brick requirement (5.1) and the projective presentations in condition Q2. Consider replacing condition Q1 by

$$\mathrm{Hom}_{\mathcal{D}_{\mathrm{fd}}(A)}(P_w, \Delta_v[k]) = 0 \quad (v <_{\sigma} w, \ 0 \leq k < m). \quad (\mathbf{P+})$$

The ambient category in (P+) is essential:  $\mathrm{Hom}_{\mathcal{H}_A^m}(P_w, \Delta_v[k])$  is defined only when  $\Delta_v[k] \in \mathcal{H}_A^m$ . For every  $v, w$  and  $k \in \mathbb{Z}$ , however,

$$\mathrm{Hom}_{\mathcal{D}_{\mathrm{fd}}(A)}(P_w, \Delta_v[k]) \cong H^k(\Delta_v)e_w. \quad (10.1)$$

Since  $\Delta_v \in \mathcal{H}_A^m \subseteq \mathcal{D}_{\mathrm{fd}}(A)^{\leq 0}$ , the terms in (10.1) vanish automatically for  $k > 0$ . Hence

$$(\mathbf{P+}) \iff H^0(\Delta_v)e_w = 0 \quad (v <_{\sigma} w). \quad (10.2)$$

Restricting (P+) to those  $k$  for which  $\Delta_v[k] \in \mathcal{H}_A^m$  gives the same condition (10.2), since  $k = 0$  is always admissible.

For  $m = 1$ , the standards are  $H^0(A)$ -modules, and (10.2) says that the composition factors of  $\Delta_v$  have weights  $u \leq_{\sigma} v$ . Together with (5.1) and condition Q2, these are classical quasi-hereditary axioms. For arbitrary  $m$ , Proposition 5.1 gives

$$\Delta_v \in \mathrm{per} A \quad (1 \leq v \leq n), \quad \mathrm{thick}_{\mathcal{D}_{\mathrm{fd}}(A)}(\Delta_1, \dots, \Delta_n) = \mathrm{per} A. \quad (10.3)$$

Indeed, descending induction in  $\leq_{\sigma}$  proves perfection from  $\Delta_{\sigma(n)} \cong P_{\sigma(n)}$ , while the projective presentations express every  $P_v$  in the thick subcategory generated by the standards.

**Example 10.1** (Two failures of standard orthogonality). For  $d \in \{1, 2\}$ , let

$$A_d = \mathbb{k}(1 \xrightarrow{a} 3 \xleftarrow{b} 2), \quad |a| = -d, \quad |b| = -1, \quad d_{A_d} = 0, \quad m = d + 1. \quad (10.4)$$

Use right modules, so  $e_1 a = a = a e_3$  and  $e_2 b = b = b e_3$ . Take  $\sigma = \mathrm{id}$  and

$$(\Delta_1, \Delta_2, \Delta_3) = (P_1, S_2, P_3), \quad (K_1, K_2, K_3) = (0, P_3[1], 0). \quad (10.5)$$

The presentations for vertices 1 and 3 are identities. For vertex 2, the short exact sequence of dg modules

$$0 \longrightarrow P_3[1] \xrightarrow{\iota_b} P_2 \xrightarrow{p_2} S_2 \longrightarrow 0, \quad \iota_b(e_3) = b, \quad (10.6)$$

has all terms in  $\mathcal{H}_{A_d}^{d+1}$ ; here  $p_2$  is the quotient map. Thus condition Q2 holds, with the single allowed factor  $K_2 = \Delta_3[1]$ . Each  $\Delta_i$  is exceptional, and  $H^0(\Delta_i) = S_i$  for  $i \in \{1, 2, 3\}$ , so both (5.1) and (P+) hold.

For every  $t \in \mathbb{Z}$ , the backward derived Hom spaces are

$$\begin{aligned} \mathrm{Hom}_{\mathcal{D}_{\mathrm{fd}}(A_d)}(\Delta_2, \Delta_1[t]) &\cong \begin{cases} \mathbb{k}, & t = 2 - d, \\ 0, & t \neq 2 - d, \end{cases} \\ \mathrm{Hom}_{\mathcal{D}_{\mathrm{fd}}(A_d)}(\Delta_3, \Delta_1[t]) &\cong \begin{cases} \mathbb{k}, & t = -d, \\ 0, & t \neq -d, \end{cases} \\ \mathrm{Hom}_{\mathcal{D}_{\mathrm{fd}}(A_d)}(\Delta_3, \Delta_2[t]) &= 0. \end{aligned} \quad (10.7)$$

The second and third equalities follow from (10.1). For the first equality,  $\mathrm{Hom}_{\mathcal{D}_{\mathrm{fd}}(A_d)}(P_2, P_1[t]) = 0$  for every  $t$ , and the triangle associated with (10.6) gives

$$\mathrm{Hom}_{\mathcal{D}_{\mathrm{fd}}(A_d)}(S_2, P_1[t]) \cong \mathrm{Hom}_{\mathcal{D}_{\mathrm{fd}}(A_d)}(P_3[1], P_1[t - 1]) \cong H^{t-2}(P_1)e_3. \quad (10.8)$$

Consequently:

- (i)  $d = 1$ : backward degree-zero orthogonality holds, but  $\text{Hom}_{\mathcal{D}_{\text{fd}}(A_1)}(\Delta_2, \Delta_1[1]) \cong \mathbb{k}$ ;
- (ii)  $d = 2$ : even backward degree-zero orthogonality fails, since  $\text{Hom}_{\mathcal{D}_{\text{fd}}(A_2)}(\Delta_2, \Delta_1) \cong \mathbb{k}$ .

Both choices also violate backward negative-degree orthogonality by (10.7).

**Example 10.2** (Membership imposed only for ordered pairs). Let

$$R = \mathbb{k}[\varepsilon]/(\varepsilon^2), \quad |\varepsilon| = -1, \quad d_R = 0, \quad A = \mathbb{k} \times R, \quad (10.9)$$

and take  $m = 2$ ,  $\sigma = \text{id}$ ,  $\Delta_i = P_i$ , and  $K_i = 0$  for  $i \in \{1, 2\}$ . Each standard is a brick, and (P+) holds. Suppose additionally that  $\Delta_v[k] \in \mathcal{H}_A^2$  is required for the same pairs and shifts as in (P+). The only pair is  $(v, w) = (1, 2)$ , and both  $\Delta_1$  and  $\Delta_1[1]$  belong to  $\mathcal{H}_A^2$ . This pairwise membership requirement places no restriction on  $\Delta_2[1]$ , which does not belong to  $\mathcal{H}_A^2$ .

The second direct factor of (10.9) is the algebra in Example 5.4. Consequently,  $A$  is not homologically smooth,  $\text{Hom}_{\mathcal{D}_{\text{fd}}(A)}(\Delta_2, \Delta_2[-1]) \cong \mathbb{k}$ , and

$$\text{thick}_{\mathcal{D}_{\text{fd}}(A)}(\Delta_1, \Delta_2) = \text{per } A \subsetneq \mathcal{D}_{\text{fd}}(A). \quad (10.10)$$

| Property under (P+)                        | Result with (5.1) and condition Q2              |
|--------------------------------------------|-------------------------------------------------|
| Classical specialisation at $m = 1$        | Yes: (10.2) is the classical support condition. |
| Generation of $\text{per } A$              | Yes, by (10.3).                                 |
| Backward degree-zero orthogonality         | Not forced: Example 10.1, $d = 2$ .             |
| Backward positive-degree orthogonality     | Not forced: Example 10.1, $d = 1$ .             |
| Exceptionality of individual standards     | Not forced: Example 10.2.                       |
| Homological smoothness                     | Not forced: Example 10.2.                       |
| Generation of $\mathcal{D}_{\text{fd}}(A)$ | Not forced: (10.10).                            |

## 11 Requiring every shifted standard to remain in the extended category

Retain (5.1), condition Q2, and (P+), and additionally require shift membership for every standard, including the greatest weight:

$$\Delta_v[k] \in \mathcal{H}_A^m \quad (1 \leq v \leq n, 0 \leq k < m) \iff \Delta_v \in \text{mod } H^0(A) \quad (1 \leq v \leq n). \quad (11.1)$$

Equation (11.1) is not part of (P+). With (11.1), (10.2) controls all cohomology of each standard. Descending induction using condition Q2 then gives (1.9); the same projective presentations give

$$\text{Hom}_{\mathcal{D}_{\text{fd}}(A)}(\Delta_v, \Delta_v[t]) \cong H^t(\Delta_v)e_v \quad (1 \leq v \leq n, t \in \mathbb{Z}). \quad (11.2)$$

The heart condition in (11.1) and the brick requirement (5.1) make each standard exceptional. Thus imposing (11.1) restricts the standards to ordinary  $H^0(A)$ -modules and recovers condition Q1.

**Example 11.1** (An excluded order on a graded arrow). Let  $A = \mathbb{k}(1 \xrightarrow{a} 2)$ , with  $e_1 a = a = a e_2$ ,  $|a| = -1$ ,  $d_A = 0$ , and  $m = 2$ . For the order  $2 <_\sigma 1$ , condition Q2 forces  $K_1 = 0$  and  $\Delta_1 \cong P_1$ . Since  $H^{-1}(P_1) = S_2$ , we have  $\Delta_1[1] \notin \mathcal{H}_A^2$ . No choice of standards for this order can satisfy (11.1). The presentation  $\Delta_i = P_i$ ,  $K_i = 0$  satisfies (C), so the extra membership condition excludes an order admitted by the exceptional-standard definition.

For the opposite order  $1 <_\sigma 2$ , the standards  $\Delta_1 = S_1$ ,  $\Delta_2 = S_2$  and kernels  $K_1 = S_2[1]$ ,  $K_2 = 0$  satisfy (11.1), (P+), and condition Q2. Thus this stronger condition restricts orders without requiring  $A$  itself to be concentrated in degree zero.



The dg dual numbers of Example 5.4 are excluded because the unique standard is  $R$  and  $R[1] \notin \mathcal{H}_R^2$ . No non-smooth algebra satisfies the requirements of Section 11: exceptionality follows from (11.2), and Corollary 2.2 applies.

| Property                                                   | Result with (11.1), (P+), (5.1), and condition Q2 |
|------------------------------------------------------------|---------------------------------------------------|
| Classical specialisation                                   | Yes; shift membership adds nothing for $m = 1$ .  |
| Backward orthogonality; exceptional terms                  | Both are consequences.                            |
| Generation of $\text{per } A = \mathcal{D}_{\text{fd}}(A)$ | Yes.                                              |
| Smoothness as a consequence                                | Yes; Corollary 2.2.                               |
| Negative cohomology of standards                           | Forbidden at every vertex.                        |
| Orders on graded acyclic path algebras                     | Some are excluded; Example 11.1.                  |
| Non-smooth examples                                        | None satisfy these requirements.                  |

## 12 Projective tests detecting negative cohomology

Retain the brick condition (5.1), the shifted-filtration axiom Q2, and  $\Delta_v \in \mathcal{H}_A^m$  for every vertex  $v$ . For  $v, w \in \{1, \dots, n\}$ , consider the following *projective tests* (P-):

$$\begin{aligned} \text{Hom}_{\mathcal{D}_{\text{fd}}(A)}(P_w, \Delta_v[-s]) &= 0 \quad (v <_{\sigma} w, 0 \leq s < m), \\ \text{Hom}_{\mathcal{D}_{\text{fd}}(A)}(P_v, \Delta_v[-s]) &= 0 \quad (1 \leq s < m). \end{aligned} \tag{P-}$$

Since

$$\text{Hom}_{\mathcal{D}_{\text{fd}}(A)}(P_w, \Delta_v[-s]) \cong H^{-s}(\Delta_v)e_w, \tag{12.1}$$

the first line of (P-) requires every  $H^t(\Delta_v)$  to have composition factors labelled  $\leq_{\sigma} v$ . The second line excludes  $S_v$  from  $H^t(\Delta_v)$  for  $t < 0$ . Under the first line, Proposition 5.3 gives

$$H^0(\Delta_v)e_v \cong \text{Hom}_{\mathcal{D}_{\text{fd}}(A)}(\Delta_v, \Delta_v) \cong \mathbb{k},$$

where the last isomorphism uses (5.1). Consequently (P-) is equivalent to (C) under the retained axioms. Proposition 7.1 therefore gives the exceptional sequence condition, fullness in  $\mathcal{D}_{\text{fd}}(A) = \text{per } A$ , and smoothness as consequences. For  $m = 1$ , the second line of (P-) is empty, and the first line is an ordinary Hom-vanishing condition for modules.

The tests (P-) also admit a formulation using only Hom spaces in  $\mathcal{H}_A^m$ . For every vertex  $w$  and  $0 \leq s < m$ , set

$$U_{w,s} := \tau^{\geq 1-m}(P_w[s]), \quad L_{w,s} := \tau^{\leq -m}(P_w[s]). \tag{12.2}$$

The amplitude of  $P_w[s]$  implies  $U_{w,s} \in \mathcal{H}_A^m$ . The truncation triangle is

$$L_{w,s} \longrightarrow P_w[s] \xrightarrow{q_{w,s}} U_{w,s} \longrightarrow L_{w,s}[1]. \tag{12.3}$$

For  $r \in \{0, 1\}$ , we have  $L_{w,s}[r] \in \mathcal{D}_{\text{fd}}(A)^{\leq -m}$  and  $\Delta_v \in \mathcal{D}_{\text{fd}}(A)^{\geq 1-m}$ . Thus

$$\text{Hom}_{\mathcal{D}_{\text{fd}}(A)}(L_{w,s}[r], \Delta_v) = 0 \quad (r \in \{0, 1\}).$$

Applying  $\text{Hom}_{\mathcal{D}_{\text{fd}}(A)}(-, \Delta_v)$  to (12.3) gives the natural isomorphisms

$$\text{Hom}_{\mathcal{H}_A^m}(U_{w,s}, \Delta_v) \xrightarrow[\sim]{q_{w,s}^*} \text{Hom}_{\mathcal{D}_{\text{fd}}(A)}(P_w[s], \Delta_v) \cong H^{-s}(\Delta_v)e_w. \tag{12.4}$$

Thus (P-) can equivalently be imposed on  $\text{Hom}_{\mathcal{H}_A^m}(U_{w,s}, \Delta_v)$  with exactly the same ranges of  $v, w, s$ . The truncations in (12.2) are necessary in this formulation:  $P_w[s]$  need not belong to  $\mathcal{H}_A^m$  when  $s > 0$ .

**Example 12.1** (An excluded one-vertex algebra). Let  $m = 2$ , let  $R = \mathbb{k}[\varepsilon]/(\varepsilon^2)$  with  $|\varepsilon| = -1$  and  $d_R = 0$ , and take  $\Delta_1 = P_1 = R$  and  $K_1 = 0$ . The brick condition and shifted-filtration axiom hold. The first line of (P-) is vacuous, whereas

$$U_{1,1} = \tau^{\geq -1}(R[1]) \simeq \mathbb{k}[1], \quad \text{Hom}_{\mathcal{H}_R^2}(U_{1,1}, R) \simeq H^{-1}(R) \simeq \mathbb{k}.$$

Here  $\mathbb{k}$  is the right  $R$ -module defined by the augmentation  $R \rightarrow \mathbb{k}$ . Thus the second line of (P-) excludes  $R$ ; imposing only the first line would still admit  $R$ .

**Example 12.2** (Standards with negative cohomology). Let  $A = \mathbb{k}Q$  for  $Q = (1 \xrightarrow{a} 2)$ , with  $|a| = -1$  and  $d_A = 0$ . Take  $m = 2$ ,  $2 <_\sigma 1$ , and  $\Delta_1 = P_1$ ,  $\Delta_2 = P_2$ . Both presentation kernels are zero, and

$$H^0(\Delta_1) \simeq S_1, \quad H^{-1}(\Delta_1) \simeq S_2, \quad H^0(\Delta_2) \simeq S_2.$$

All other cohomology groups vanish. Therefore (P-) holds, and the standard sequence is  $(\Delta_2, \Delta_1) = (P_2, P_1)$ . In particular, the projective tests permit negative cohomology at strictly smaller vertices and do not require the standards to lie in the heart  $\text{mod } H^0(A)$ .

| Property                     | Consequence of (P-), (5.1), and Q2                                        |
|------------------------------|---------------------------------------------------------------------------|
| Specialisation at $m = 1$    | The classical quasi-hereditary definition.                                |
| Backward orthogonality       | Vanishing in every degree.                                                |
| Self-orthogonality           | Every $\Delta_v$ is exceptional.                                          |
| Fullness                     | $\Delta^\sigma$ is full in $\mathcal{D}_{\text{fd}}(A) = \text{per } A$ . |
| Smoothness                   | A consequence of Proposition 7.1.                                         |
| Negative standard cohomology | Allowed at vertices strictly smaller than the label of the standard.      |
| Non-smooth counterexamples   | None, by Proposition 7.1.                                                 |

## 13 Immediate problems

**U1. Compare separate and combined amplitude bounds.** Suppose a full exceptional sequence has presentations (1.10) with  $K_v \in \mathcal{Q}_v$ ,  $\Delta_v \in \mathcal{Q}_v^\perp$ ,  $\Delta_v \in \mathcal{H}_A^m$ , and  $\nabla_v \in \mathcal{H}_A^m[1-m]$ . The two amplitude conditions separately give  $\alpha_v \leq m-1$  and  $\beta_v \leq m-1$ . Determine whether there are such examples with  $\alpha_v + \beta_v > m-1$  for some  $v$ . An example would separate these conditions from Definition 1.1 at the same width. The degreewise obstruction is precisely the non-vanishing in (4.3) outside the allowed shifts.

**U2. Recognise the canonical standards from the dga.** Given a basic proper connective dga  $A$  and an order  $\sigma$ , determine when every  $P_v$  admits a triangle  $K_v \rightarrow P_v \rightarrow \Delta_v \rightarrow K_v[1]$  with  $K_v \in \mathcal{Q}_v$  and  $\Delta_v \in \mathcal{Q}_v^\perp$ , where  $\Delta_v$  is exceptional and lies in  $\mathcal{D}_{\text{fd}}(A)^{\leq 0}$ . Seek criteria on the dg ideals  $Ae_{>v}^\sigma A$  and their gluing bimodules. Once the exceptional standard and left-dual objects are constructed, Proposition 4.1 tests the filtration axiom, and Corollary 4.2 determines the width whenever one width exists.

## 14 Further problems

**P1. Standardisation.** Starting from an ordered exceptional family and allowed shifted filtrations in an extriangulated category, determine the conditions for a dg algebra to realise Definition 1.1. Specify the necessary enhancement and higher-composition data. Compare with [AT24], checking its negative-extension hypotheses.

**P2. Ringel duality.** Determine when the categories generated by allowed shifted standards and by the shifted costandards in (4.10) admit a common tilting or silting object  $T \in \mathcal{D}_{\text{fd}}(A)$ . For the derived endomorphism dga  $\mathbf{R}\text{End}_A(T)$ , determine connectivity, the induced ordering, the width, and the effect of iterating the construction.

**P3. Homological bounds.** Seek bounds depending on  $m$  and  $n$  for lengths of standard resolutions and for

$$\sup\{t \in \mathbb{Z} : \operatorname{Hom}_{\mathcal{D}_{\text{fd}}(A)}(S_u, S_v[t]) \neq 0 \text{ for some } 1 \leq u, v \leq n\}.$$

**P4. Heredity ideals and degree-zero reduction.** Relate (2.7) to derived idempotent quotients and chains of dg ideals. Determine the additional conditions on the gluing data that encode (1.11), in comparison with the classical recollement characterisation [Kra17]. Although (2.8) identifies  $H^0(\Delta_v)$  with the ordinary standard module, Example 3.4 shows that  $H^0(A)$  need not be quasi-hereditary for the same ordering. Characterise when quasi-heredity descends to  $H^0(A)$ .

**P5. Changes of order and equivalence.** Determine which mutations of  $\Delta^\sigma$  preserve (1.10) and (1.11) at a fixed width. Study how the standard objects and the least-width formula (4.8) change under silting mutation of the projective generator and under changes of total order on the fixed vertex labels.

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